温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W03
Strategic Mgt Global Info Sys

Introduction to MIS

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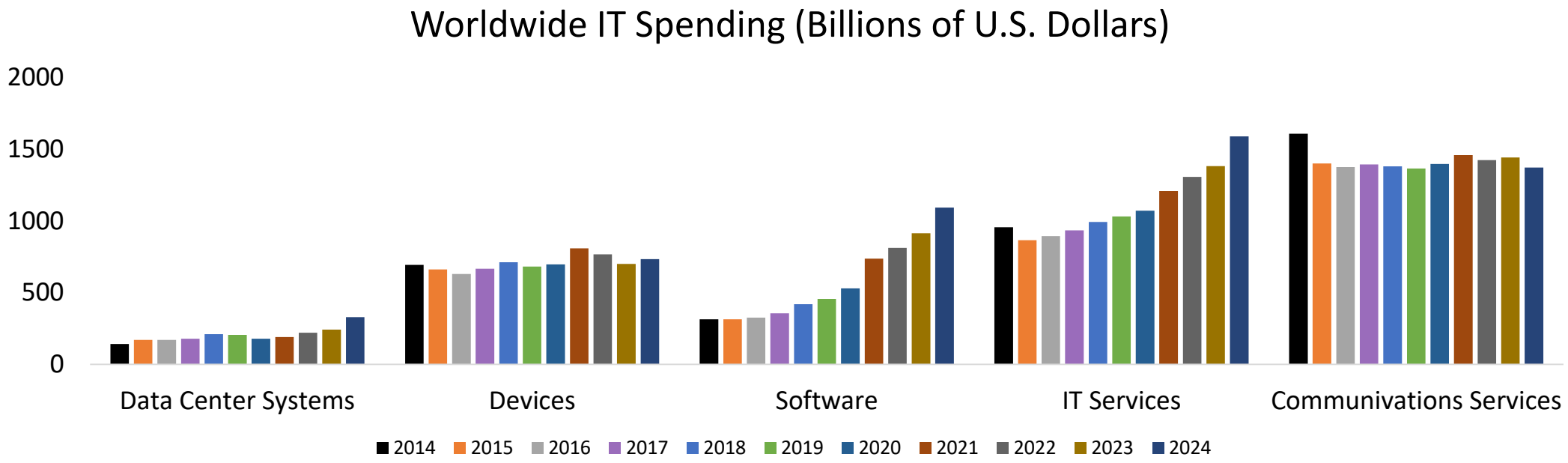
Learning Objectives

By the end of this session, you should be able to:

- Explain how information systems transform businesses.
- Identify the business objectives supported by information systems.
- **Define** an information system and describe how it works.
- Distinguish the organizational, management, and technology **dimensions** of IS.
- Recognize key challenges and opportunities from emerging MIS technologies.
- Compare the different approaches to studying information systems.

Why MIS Matters



Source: Gartner

If a business isn't using information systems connected to the Internet and mobile apps, chances are it **isn't being as effective as it could be**.

Why MIS Matters

- In today's digital economy, *data is as critical as capital or labor*.
- MIS provides structure and intelligence to *how data becomes value*.
- Technology-driven changes are transforming jobs and careers:
  - How you work, where you work, and how well you will be compensated
- Many millions of new jobs have been created
  - Not only for IT specialists but also for business professionals who use IT tools
- Gaining a deeper understanding of how technology works and is evolving is essential (e.g., AI tools).

"Every global company is becoming a tech company … we're seeing technology as a critical component for business success."

– Jeremy King, CTO@Walmart Labs

Why MIS Matters

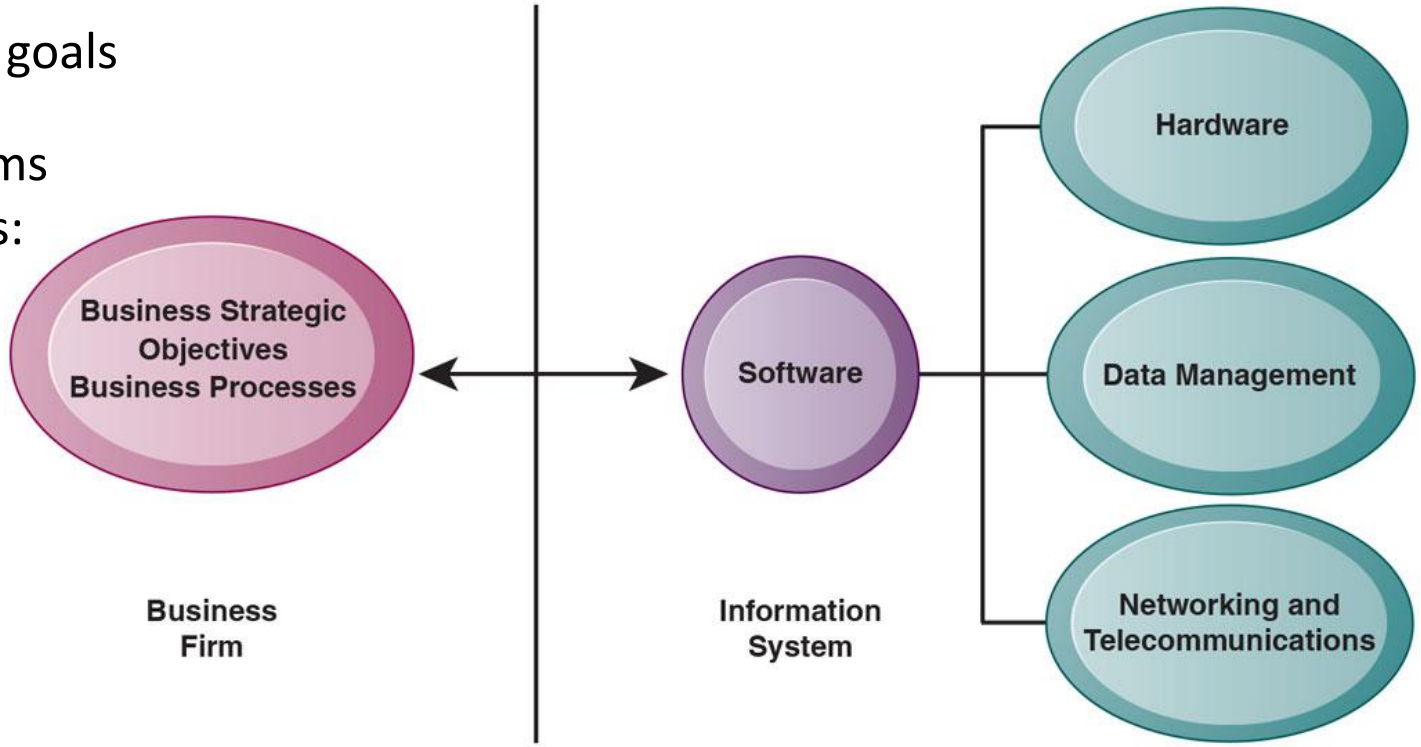
Growing interdependence between

- Ability to use information technology
- Ability to implement corporate strategies and achieve corporate goals

Firms invest heavily in information systems to achieve 6 strategic business objectives:

- Operational excellence
- New products, services, and business models
- Customer and supplier intimacy
- Improved decision making
- Competitive advantage
- Survival

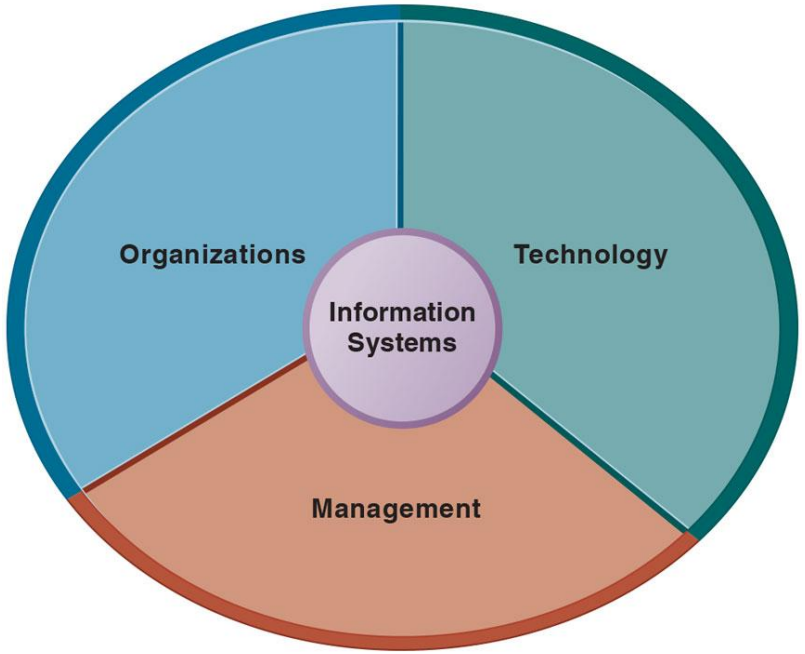
Interdependence of Organization and Information Systems



Source: Laudon Ch1

What Is MIS

- Information Technology (IT)
  - The hardware and software a business uses to achieve objectives
- Management Information System (MIS/IS)
  - A set of interrelated components that **collect** (or retrieve), **process**, **store**, and **distribute** information to
    - (a) Support decision making, coordinating and control
    - (b) Help analyze problem, visualize complex subjects, and create new products
- The study of MIS deals with behavioral and technical issues surrounding the **development**, **use**, and **impact** of information systems that managers and employees in the firm use.



Source: Laudon Ch1

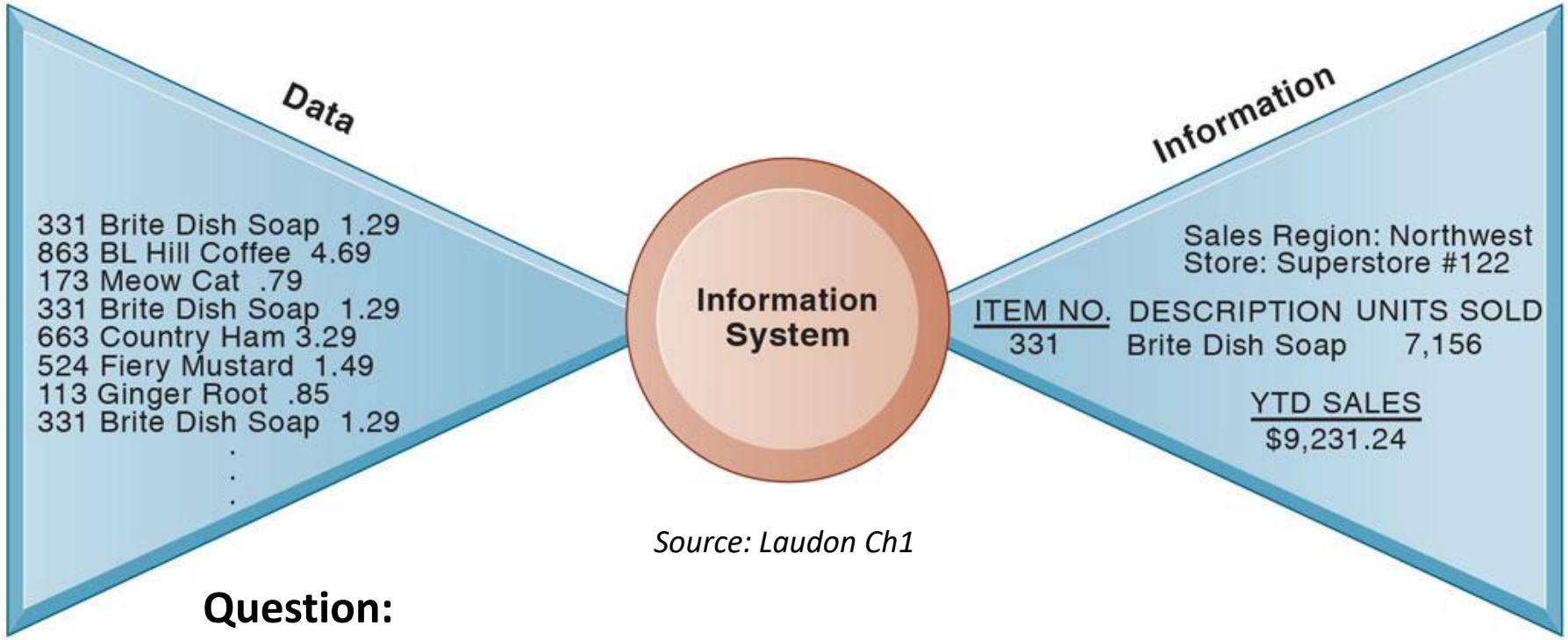
What Is MIS

Data, Information, and Knowledge

	Data	Information	Knowledge
Definition	Simple observations of the state of the world	Data endowed with relevance and purpose	Information from the human mind (includes reflection, synthesis, context)
Characteristics	• Easily structured • Easily captured on machines • Often quantified • Easily transferred • Mere facts	• Requires unit of analysis • Data that have been processed • Human mediation necessary	• Hard to structure • Difficult to capture on machines • Often tacit • Hard to transfer
Example	Daily inventory report of all items	Daily inventory report of items with low levels of stock	Report of which items need to be reordered after considering inventory levels, anticipated labor strikes, and a flood affecting a supplier

What Is MIS

Example: Raw data from a *supermarket checkout counter* can be processed and organized to produce meaningful information.



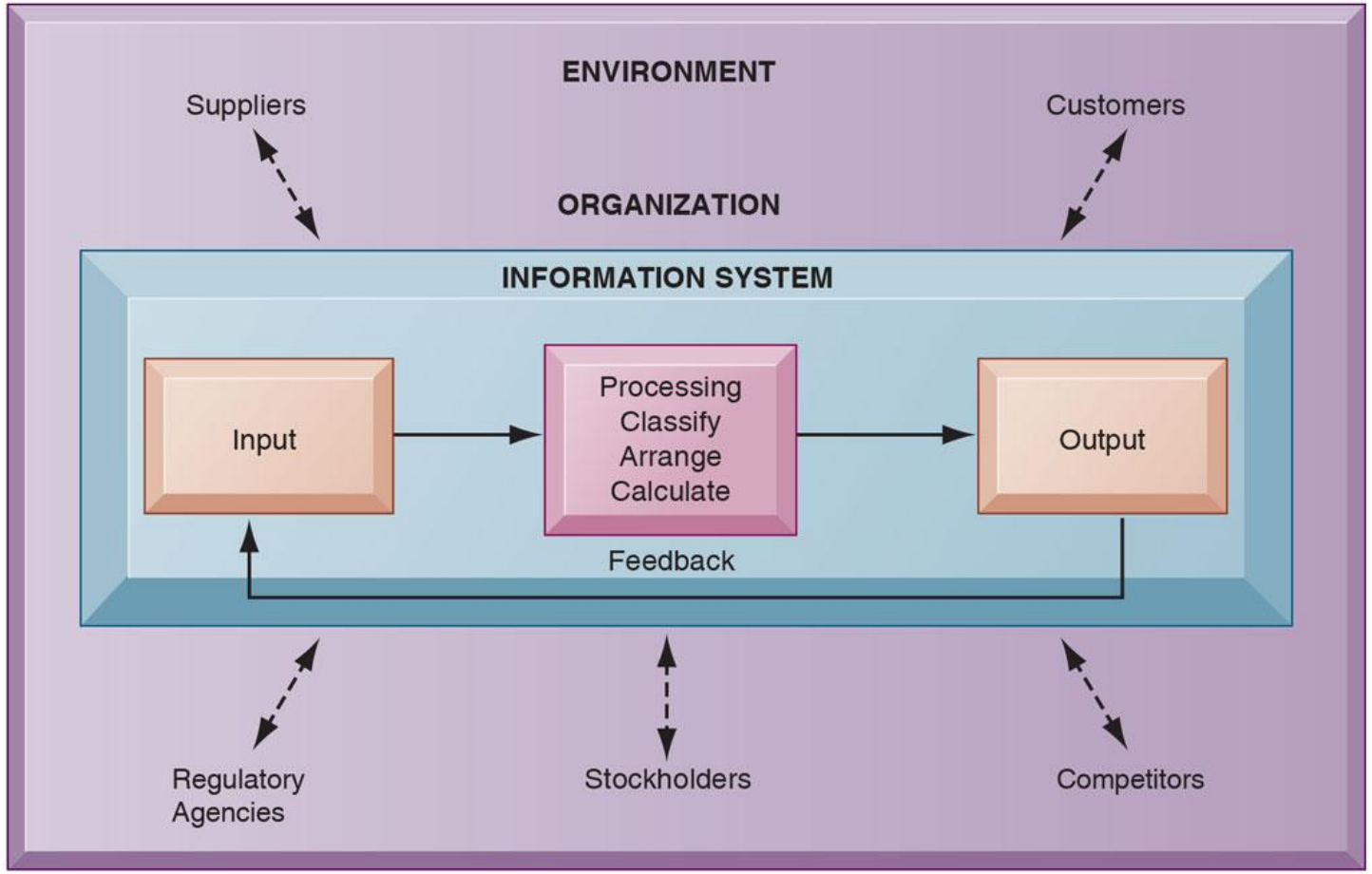
Source: Laudon Ch1

Question:

- What are data in this example?
- What are information in this example?
- Can you provide one or two examples to illustrate the difference?

What Is MIS

Functions of an Information System



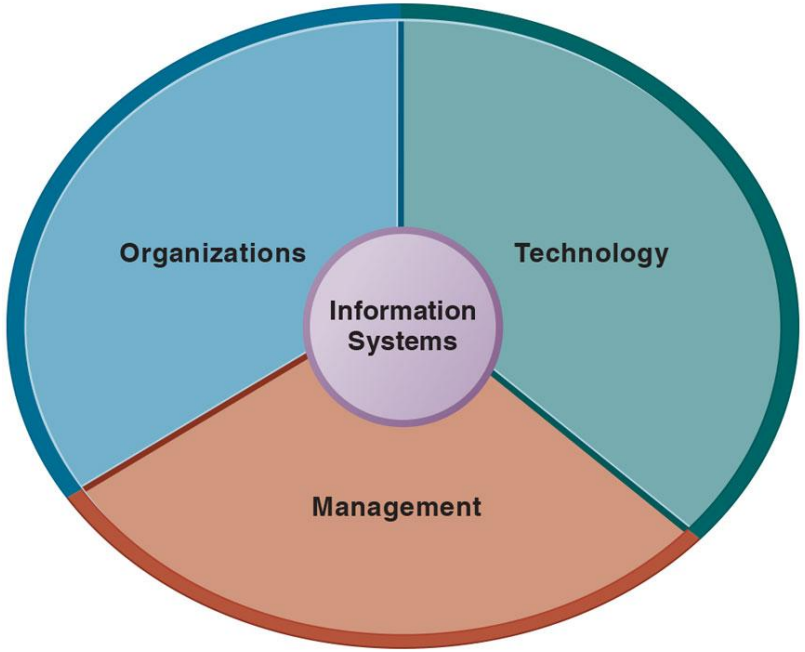
Source: Laudon Ch1

- Three core activities
  - **Input:** Captures raw data from inside or outside the organization.
  - **Processing:** Converts data into meaningful information.
  - **Output:** Distributes processed information to users and decision-makers.
- Feedback: Output returned to relevant people or systems to *improve future inputs and processes*.
- External actors: Customers, suppliers, competitors, stockholders, and regulators *continuously interact* with the system.

What Is MIS

To understand information systems fully:

- You need to be aware of the **broader** organizational, managerial, and information technology dimensions of systems
- Also, their power to provide *solutions to challenges and problems* in the business environment



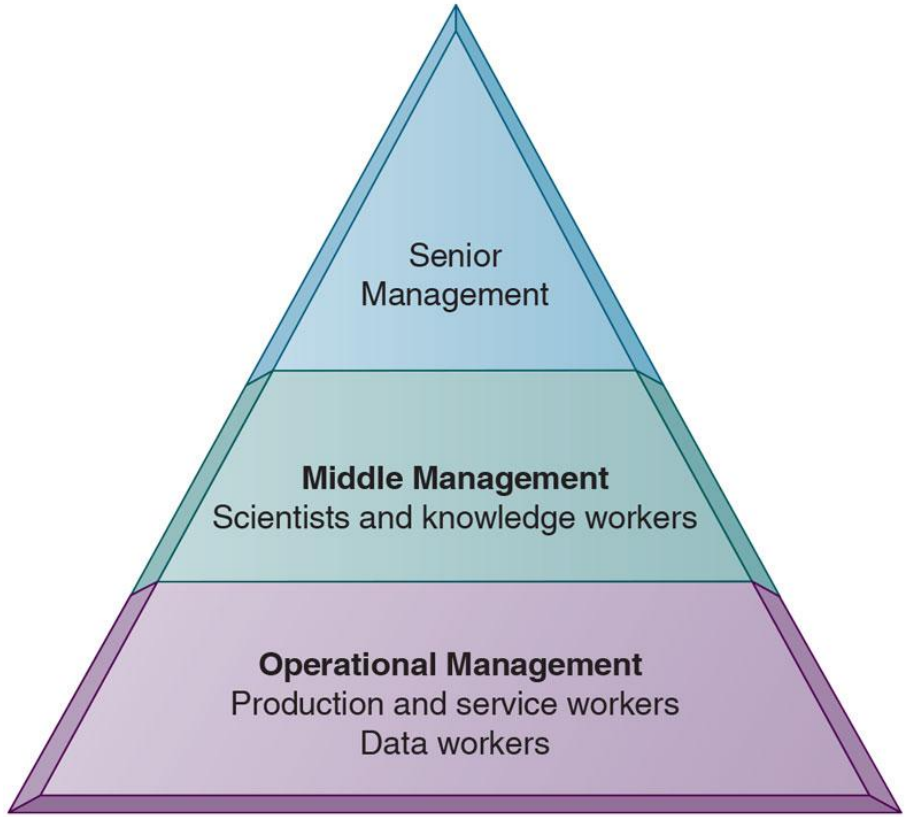
Information systems are **more than** computers.

Source: Laudon Ch1

What Is MIS – Organization

- Two-Way Relationship
  - Information systems transform how firms operate.
  - Organizational **history**, **culture**, and **structure** determine how technology is adopted and used.
- Organizational Structure
  - Defined by levels and specializations — a hierarchy of authority and responsibility.
  - Clear division of labor across managerial and operational
- Key Idea
  - To understand how a firm uses IS, analyze its **structure**, **culture**, and **managerial layers**.

Levels in a Firm



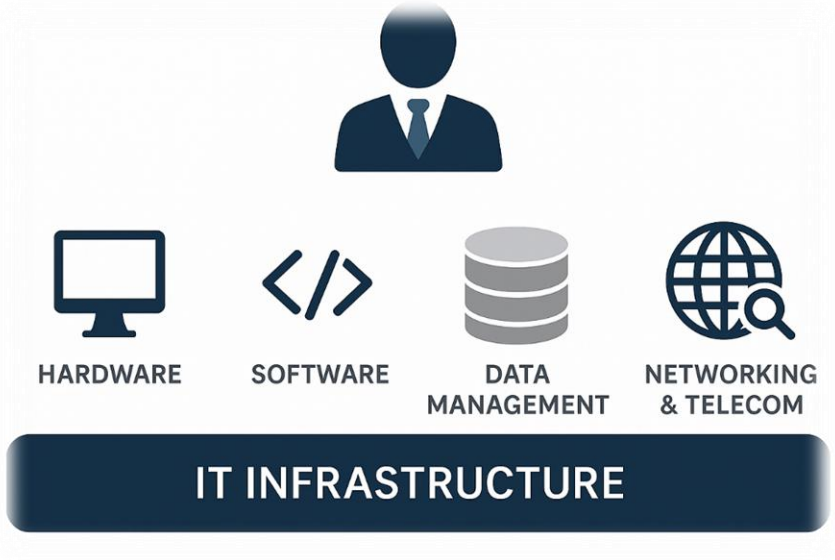
Source: Laudon Ch1

What Is MIS – Management

- Managerial Roles
  - Interpret challenges and opportunities
  - Set strategy, made decisions, allocate resources
  - Lead teams and align goals
  - Drive Innovation
- Information Systems as **Enablers**
  - Provide data, models, and analytical tools
  - Support collaboration, innovation, and evidence-based decisions
  - Convert knowledge into competitive advantage

What Is MIS – Technology

- Four Core Components
  - **Hardware:** Devices for input, processing, storage, and networking
  - **Software:** Programs that control and coordinate hardware
  - **Data Management:** Organizes, stores, and retrieves data efficiently
  - **Networking & Telecom:** Connects systems and enables data, voice, and video exchange
- IT + People → IT Infrastructure
  - The shared foundation of technology and people supporting all IS
  - Provides the platform for building and managing organizational systems



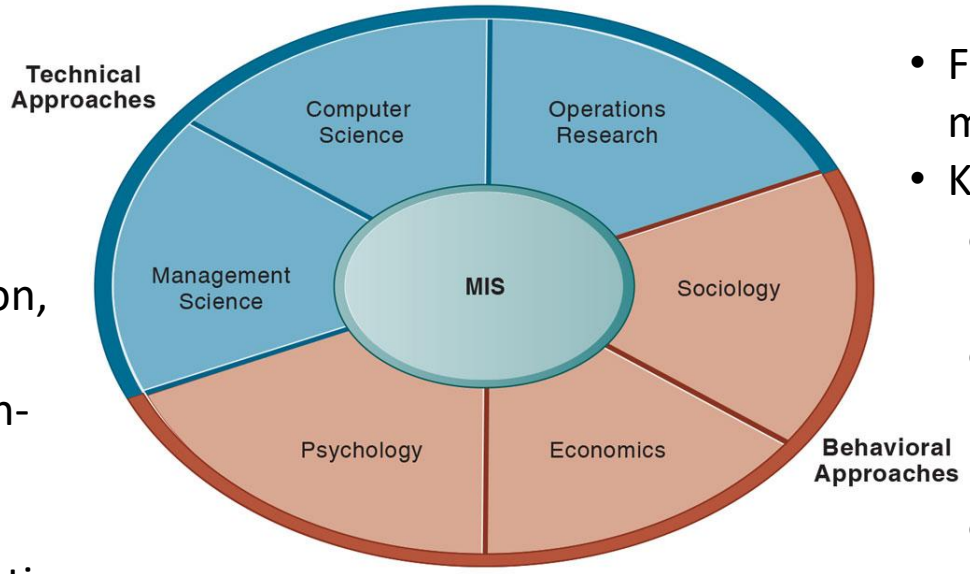
What Is MIS – Technology

- Internet of Things (IoT)
  - A network of physical objects with sensors and software
  - Connects and exchanges data via the Internet
- Cloud Computing
  - Shared pool of computing resources (storage, software, processing)
  - Accessed on demand via the Internet
- Artificial Intelligence (AI)
  - Computer systems performing tasks normally requiring human intelligence
  - Examples: Speech and image recognition, predictions, learning
- Machine Learning (ML)
  - Subfield of AI where systems learn and improve automatically
  - Uses algorithms to detect patterns in large data sets
- Generative AI
  - Identify data patterns and create new content – text, code, image, audio, video
  - Examples: ChatGPT, Gemini, Copilot

Approaches to Study MIS

- The study of information systems is a **multidisciplinary field**.
- No single theory or perspective dominates.

- Focus: Mathematical models, technologies, and formal system capabilities
- Key Disciplines
  - **Computer Science:** computation, data storage, and access
  - **Management Science:** decision-making models and analytical methods
  - **Operations Research:** optimization of logistics, inventory, and costs



Source: Laudon Ch1

- Focus: Human, organizational, and managerial dimensions of IS
- Key Disciplines
  - **Sociology:** how systems shape and are shaped by organizations
  - **Psychology:** how people perceive, use, and respond to information systems
  - **Economics/Management:** behavior, incentives, and system utilization

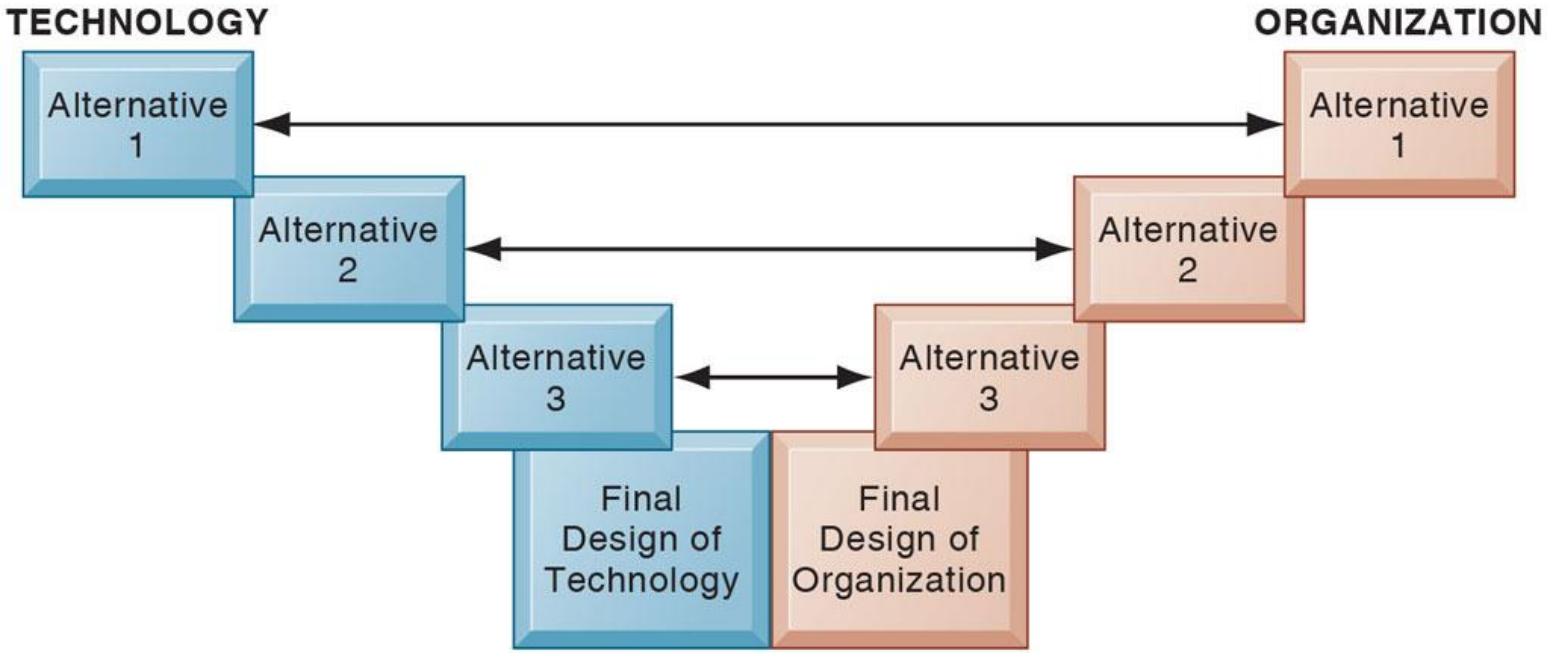
Research Topics in MIS

Research Topic in MIS (MIS Quarterly Research Curations)

- Information Systems Development
- IS Sourcing
- IT Project Management
- Data Management
- Information Systems Alignment
- IS Use
- Knowledge Management
- Health Information Technology
- IT-supported Collaboration
- Online Word-of-Mouth
- Platforms and Ecosystems
- IT Workforce
- Information Privacy
- IS Control & Governance
- Societal Implications of ICTs

A Sociotechnical Perspective on MIS

Optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production



Source: Laudon Ch1

Key Challenges in New MIS Technologies

Technology Changes	Business Challenges
Cloud Computing Organizations move IT operations to cloud-based platforms for scalability and cost efficiency.	How much IT infrastructure should move to the cloud? What are the trade-offs in cost, control, and security?
Big Data & IoT Firms collect vast data from sensors, web traffic, and social media to improve decisions.	Does your firm have the tools and talent to analyze and leverage big data effectively?
Artificial Intelligence (AI) Systems perform tasks requiring human-like intelligence—recognizing patterns, predicting, learning.	Where can your company apply AI? What benefits, costs, and expertise are needed for implementation?
Mobile Platforms Smartphones, tablets, and wearables enable collaboration and connectivity anywhere.	Is your firm maximizing mobile capabilities for employees and customers? How can it enhance value?

Source: Laudon Ch1

Key Challenges in New MIS Technologies

Management	Business Challenges
Return on Investment (ROI) Firms invest heavily in information systems but often struggle to measure the value gained.	How can your firm accurately assess the benefits of IT investments and identify more cost-effective alternatives?
Online Collaboration & Social Networks Platforms like Google Workspace, Microsoft 365, Slack, and Zoom reshape teamwork and communication.	Is your firm effectively using digital tools to improve coordination and knowledge sharing? Which platforms add the most value?

Source: Laudon Ch1

Key Challenges in New MIS Technologies

Organizations	Business Challenges
Security & Privacy Data protection and privacy have become critical public and regulatory concerns.	How secure is your company's data? Are its privacy policies adequate and compliant with evolving laws?
Social Business Firms leverage social platforms (Facebook, X, TikTok, internal networks) to engage employees and customers.	Is your firm gaining measurable value from social business tools? How could engagement be deepened?
Remote & Hybrid Work Cloud and mobile technologies enable flexible work models across locations.	What policies and technologies will shape your firm's future of work? What challenges and opportunities exist?

Source: Laudon Ch1

Check Your Understanding

Which of the following is **not** an important business objective driving investment in information systems and technologies?

A. Improved employee benefits
B. Improved decision making
C. New products, services, and business models
D. Operational excellence

Check Your Understanding

Processing involves which of the following?

A. Returning output to appropriate members of the organization to help them evaluate or correct the input stage.
B. Transferring information to the people who will use it or to the activities for which it will be used.
C. Capturing or collecting raw data from within the organization or from its external environment.
D. Converting data into a meaningful form by classifying, arranging, and performing calculations on it.

Check Your Understanding

Which of the following is especially useful for finding patterns in very large amounts of data and predicting outcomes?

A. Cloud computing
B. Extranets
C. Internet of things
D. Machine learning

Summary

- MIS enables firms to achieve efficiency, innovation, and competitiveness.
- Data, information, and knowledge
- MIS is a set of interrelated components that collect (or retrieve), process, store, and distribute information.
  - Three dimensions: organization, management, technology
- Approaches to studying MIS
  - Technical, behavioral, sociotechnical
- Key challenges in new MIS technologies
  - Aligning IT with business strategy
  - Ensuring data security, privacy, and ethical use
  - Managing digital transformation and rapid technological change